

**Modern College Of Art, Commerce And Science
Ganeshkhind, Pune-16 (Autonomous)**



TYBSc.Computer Science

ACADEMIC YEAR 2024-2025

“Hostel Harmony System”

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P.E.Society's
MODERN COLLEGE OF ARTS, SCIENCE AND COMMERCE,
GANESHKHIND, PUNE-16 (Autonomous)



Certificate

This is to certify that project titled “Hostel Harmony System” has been successfully completed by Shree Nigade (2433321078) , Ashish Ranmale (2433321094) and Mansi Kawale(2433321051) as fulfilment of T.Y.B.sc (Computer science) Examination for the academic year 2024-2025 for Savitribai Phule Pune University.

Project Guide

Examiner

Head of Dept.

Date :

College Seal:

PREFACE

It gives me great pleasure to present the project titled “Hostel Harmony System,” prepared with my utmost sincerity and dedication. The content of this document is systematically organized into multiple chapters to ensure a smooth and comprehensive reading experience.

This project has been thoroughly illustrated with clear and accurate diagrams, including the State Diagram, ER Diagram, Use Case Diagram, and Class Diagram, all of which provide a detailed understanding of the system's design and functionality.

The *Hostel Harmony System* aims to simplify hostel management by streamlining processes such as room allocation, fee tracking, grievance handling, and communication between residents and administrators. It is a robust and user-friendly solution designed to foster an efficient and harmonious living environment.

This project reflects a practical approach to addressing the real-world challenges of hostel management and offers a significant contribution toward modernizing such systems.

ACKNOWLEDGEMENT

The satisfaction that accompanies the successful completion of any task would be incomplete without expressing our sincere gratitude to **Prerana Sarode Ma'am** for her invaluable guidance, insightful advice, and constant support. Her expertise has been instrumental in shaping the direction of this project and ensuring its successful completion.

We also extend our heartfelt appreciation to our faculty members and institution for providing us with the necessary resources and a conducive learning environment. Their encouragement and support have played a vital role in our academic growth.

Furthermore, we express our gratitude to our colleagues and teammates for their cooperation, valuable suggestions, and assistance throughout this project. Their contributions have made this project more enriching and productive.

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Abstract

The **Hostel Harmony System** is a comprehensive online platform designed to simplify hostel management for administrators and residents. It aims to eliminate inefficiencies in traditional paper-based systems by providing a centralized digital solution for tasks such as room allocation, fee management, complaint tracking, and communication. This system is particularly relevant in modern times as hostels cater to a growing population of students and professionals who demand seamless and transparent management. By integrating advanced technology, the project ensures an efficient, user-friendly experience that aligns with the digital transformation trends in education and housing.

INTRODUCTION

The **Hostel Harmony System** is an innovative online platform designed to simplify and automate hostel management processes. It replaces traditional paper-based methods with a centralized digital solution, enabling efficient handling of tasks like room allocation, fee management, complaint resolution, and communication. The system aims to enhance operational transparency and improve the user experience for both administrators and residents, ensuring seamless and effective management in educational and professional housing facilities.

Managing hostels using traditional, manual methods often leads to inefficiencies such as misplaced records, communication gaps, and delayed issue resolution. These challenges result in dissatisfaction among residents and increased workload for administrators. The need for the **Hostel Harmony System** arises from the demand for a streamlined, digital solution that automates and centralizes hostel operations, ensuring accuracy, transparency, and improved management efficiency.

The motivation behind the **Hostel Harmony System** stems from the growing reliance on digital solutions to simplify daily operations. With the increasing number of students and professionals residing in hostels, an efficient management system is essential to meet their needs. This project is highly relevant in the real world as it enhances operational efficiency, reduces administrative burdens, and provides residents with a seamless, user-friendly experience, aligning with the global shift toward digital transformation in housing and education sectors.

Problem Statement

The **Hostel Harmony System** is an innovative online platform designed to simplify and automate hostel management processes. It replaces traditional paper-based methods with a centralized digital solution, enabling efficient handling of tasks like room allocation, fee management, complaint resolution, and communication. The system aims to enhance operational transparency and improve the user experience for both administrators and residents, ensuring seamless and effective management in educational and professional housing facilities.

Managing hostels using traditional, manual methods often leads to inefficiencies such as misplaced records, communication gaps, and delayed issue resolution. These challenges result in dissatisfaction among residents and increased workload for administrators. The need for the **Hostel Harmony System** arises from the demand for a streamlined, digital solution that automates and centralizes hostel operations, ensuring accuracy, transparency, and improved management efficiency.

Objective and Goals

Objectives:

- To create a user-friendly online platform for hostel management.
- To automate tasks such as room allocation, fee management, and complaint resolution.
- To provide secure and transparent communication channels between administrators and residents.

Goal:

The primary goal of the **Hostel Harmony System** is to enhance the efficiency and effectiveness of hostel operations by leveraging technology, thereby improving the satisfaction of both administrators and residents.

Scope And Limitations

Scope of proposed Hostel Harmony System project would encompass following key features:

- 1) User Authentication and Authorization
- 2) Admin Login for System administrators
- 3) Student/Resident login for hostel occupants .
- 4) Add, edit, delete, student profile, Assign and allocate
- 5) Allocate and deallocate rooms to student ,view and occupancy status Billing and payment management
- 6) Implement security measure like CCTV surveillance
- 7) Tracking and managing Hostel supplies and inventory
- 8) Allowing students to submit Feedbacks and Complaints
- 9) Facilitate communication between administrator and student

Limitation

- **Technical Issues:** System downtimes and compatibility problems.
- **Cost:** High setup, maintenance, and update costs.
- **Data Security:** Risks of data breaches and cyberattacks.
- **Internet Dependency:** Requires a stable internet connection.
- **Limited Customization:** May not cater to specific hostel needs.
- **Scalability:** Difficulty in scaling for large institutions.
- **Complexity:** Overly complex for smaller hostels

Existing System

- **Online Booking:** Students can book rooms via the hostel's website or app.
- **Real-time Availability:** Displays room availability, making booking easier.
- **Online Payments:** Payments are made through gateways, and receipts are mailed.
- **Database Management:** Stores student and booking details securely for easy access.
- **Automated Room Assignment:** Rooms are assigned based on availability and preferences.
- **Notifications:** Automated confirmations, reminders, and alerts are sent via email/SMS.
- **Reporting:** Generates occupancy and payment reports for hostel management.
- **Admin Panel:** Admins can manage bookings, student details, and room availability.

Limitation of existing system:

The existing Online Hostel Harmony System has limitations such as dependence on a stable internet connection, limited customization options for room preferences, and potential technical issues like system downtime or glitches. There are security concerns regarding data protection, and online payment failures can disrupt bookings. Admins and students may face challenges if they are not adequately trained on using the system. Compatibility issues with certain devices or browsers, as well as the system's inability to handle high volumes of bookings during peak times, are also drawbacks.

Proposed System:

- **Centralized Platform:** Manages student registrations, room allotments, and other hostel operations in one system.
- **Room Allotment & Fee Management:** Automates room assignments based on availability and tracks fee payments with reminders for pending dues.
- **Real-time Updates & Notifications:** Provides real-time updates on room availability and fee status, sending automatic notifications to users.
- **Reporting:** Generates reports on occupancy, fee collection, and student activity for administrators.
- **User-friendly Interface & Secure Access:** Offers an easy-to-use interface with secure login for authorized users, improving efficiency and reducing manual errors.

Project Perspective:

The Hostel Harmony System is designed as a web-based application, ensuring easy access through any modern web browser with an active internet connection. The system provides a user-friendly interface that can be easily navigated by hostel administrators, staff, and students. The graphical user interface (GUI) is designed to be intuitive, allowing users to efficiently manage tasks such as room allotment, fee payments, and student records. Once the application is loaded, users can seamlessly interact with the system to view and update data, making hostel management processes more organized and accessible. The system's design ensures it is simple to maintain, making it suitable for use by non-technical personnel as well.

Features :

- **Room Booking System:** Allow students to book rooms based on availability, room type, and preferences with an easy-to-use booking interface.
- **Student Profiles:** Enable students to create and manage personal profiles, storing details such as room preferences, fee payments, and booking history.
- **Fee Management:** Provide a feature to view, track, and pay hostel fees online, along with automated reminders for pending payments.
- **Room Allocation & Availability:** Offer real-time room availability updates and automated room allocation based on student preferences and hostel capacity.
- **Feedback and Rating System:** Allow students to rate and review hostel facilities, services, and management to help improve overall operations.
- **Complaint Management System:** Allow students to raise complaints or maintenance requests, track their status, and ensure timely resolution.
- **Admin Dashboard:** Provide hostel administrators with an intuitive dashboard to manage rooms, monitor student activities, and generate reports efficiently.

Requirement Analysis:

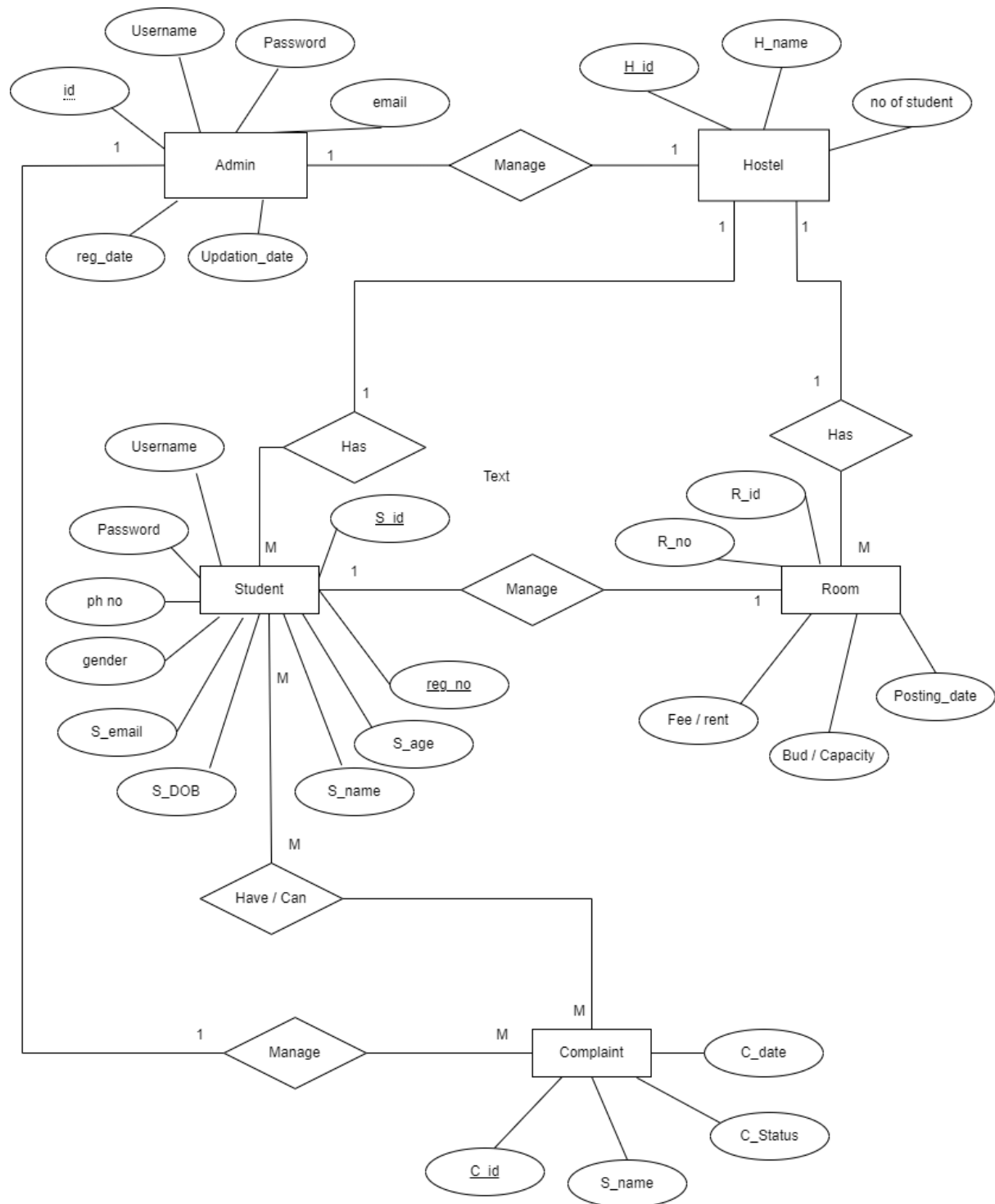
The Functional Requirements of the project are as follows:

- Should have a stable internet connection.
- Should be an authorized member to appear for the login system.
- The system should accept all the sessions.

Stakeholders:

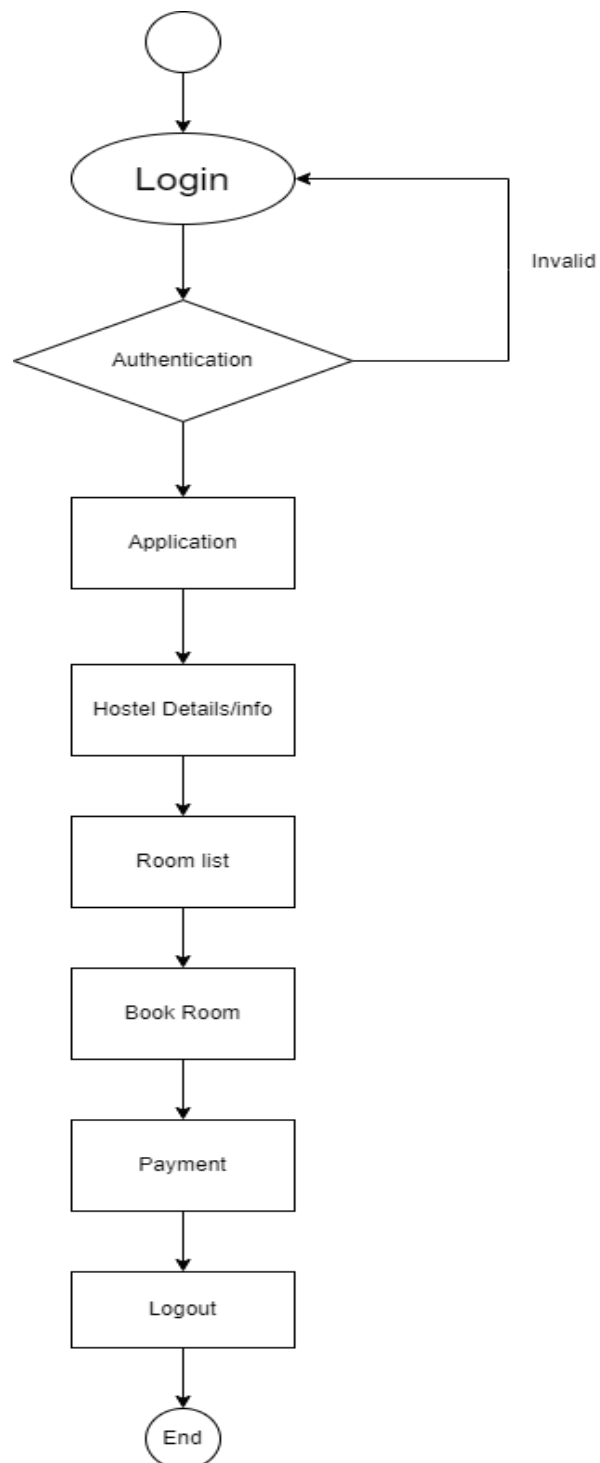
- Admin
- End Users

ER Diagram :-

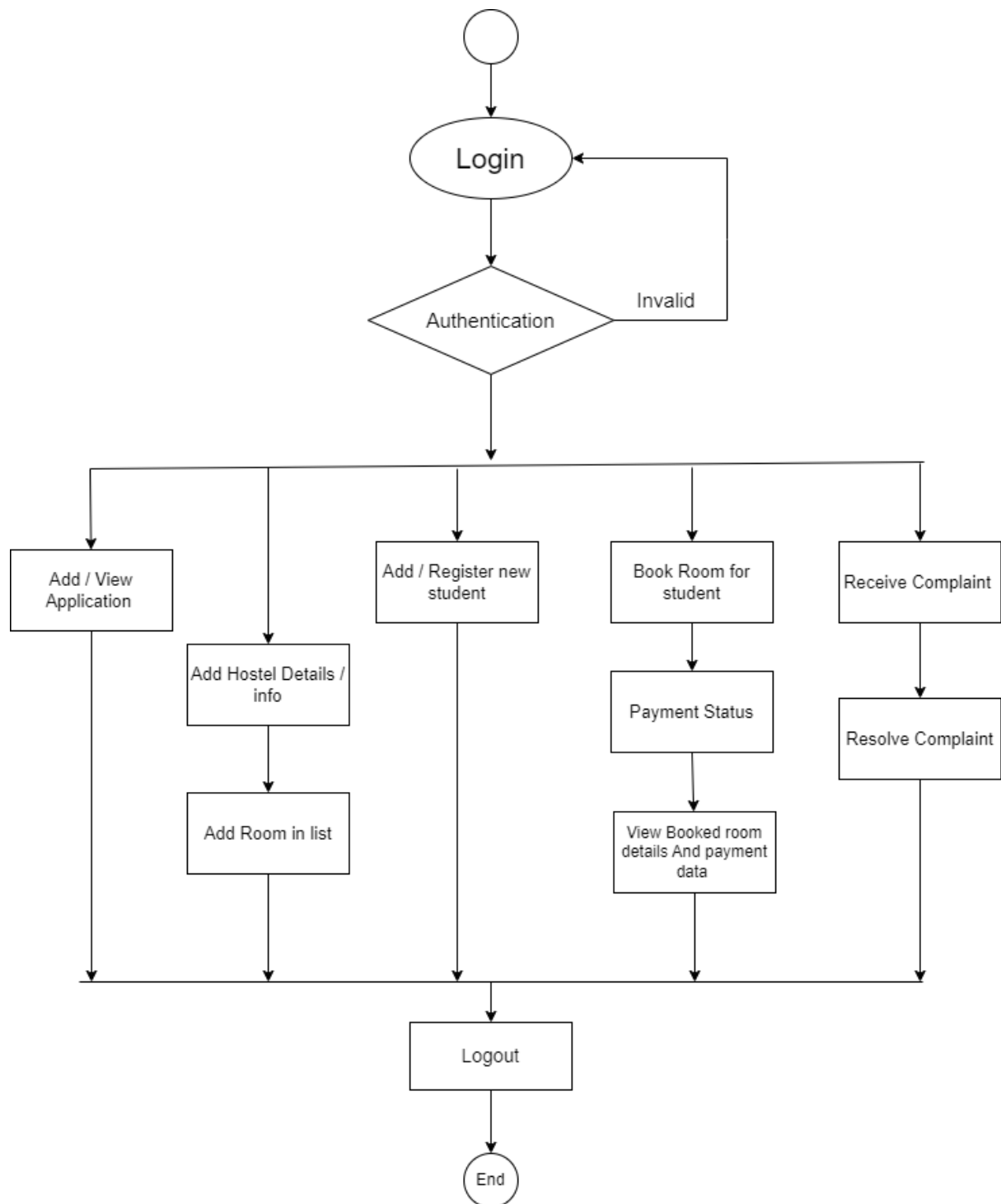


Activity Diagram :

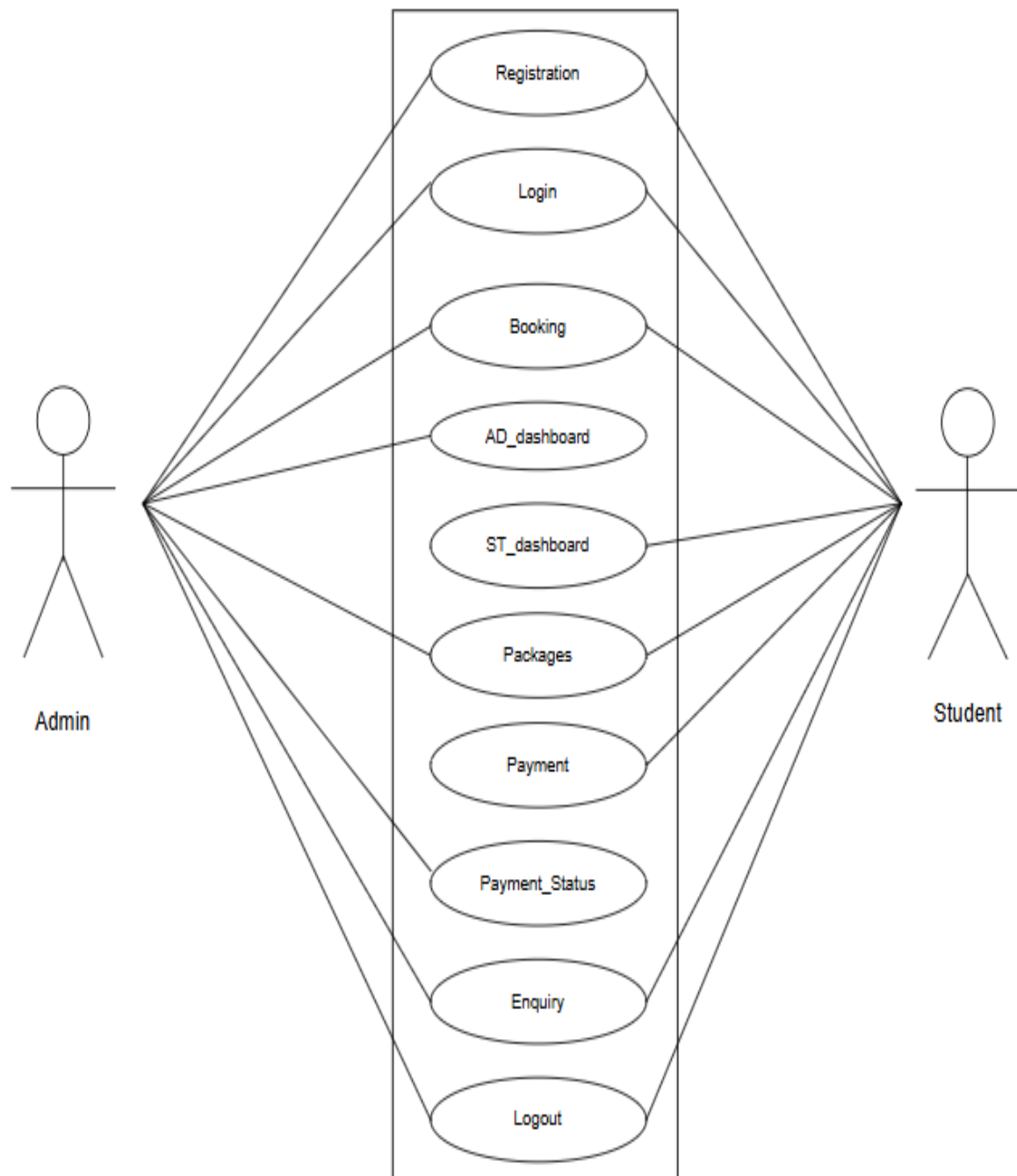
1.Student :-



2. Admin:-



Use-case Diagram:



User Module :

Student Login

Email:

Password:

Login

Admin Login



Welcome Back!

To keep connected with us please login with your personal info

SIGN IN

Create Account

Use your email for registration:

SIGN UP



HOSTEL HARMONY SYSTEM

[About Us](#)[Feedback](#)[Facilities](#)[Log In](#)[Sign Up](#)

Creating Comfort, Crafting Memories..

where comfort, safety, and a sense of community come together to offer you the ultimate living experience. Our hostel is designed with your needs in mind, providing a home-like atmosphere, modern amenities, and a vibrant space for learning, growing, and connecting with fellow students. Whether you're here for a short-term stay or a longer journey, we ensure every moment spent with us is both comfortable and memorable.*.

[Start Your Free 30-Day Trial](#)[Contact Us](#)



Welcome to our Harmony

Your Home Away from Home

About Us

Our Hostel Management System ensures a seamless experience for all residents by providing excellent facilities and services. We strive to create a welcoming environment where students can focus on their studies and personal growth. The hostel offers spacious rooms with cozy beds and study tables, ensuring a peaceful place for both rest and study. We provide fresh, nutritious meals, including both vegetarian and non-vegetarian options. 24/7 security with CCTV and on-site personnel ensures your safety, while dedicated wardens are always available to assist with any needs. The hostel also features a well-stocked library, high-speed Wi-Fi, recreational areas, and convenient laundry services. All these amenities make it the perfect place for students to live, study, and thrive.



Comfortable Rooms



Quality Beds



Delicious Food

Our hostel provides nutritious and hygienic meals catering to all dietary preferences.



Library Facility

A well-stocked library is available to residents for academic and leisure reading.



High-Speed Wi-Fi

Uninterrupted Wi-Fi connectivity for all your academic and personal needs.



24/7 Security

Round-the-clock security ensures the safety and well-being of all residents.



Dashboard

Book Room

File Complaint

My Room Details

Logout

Welcome

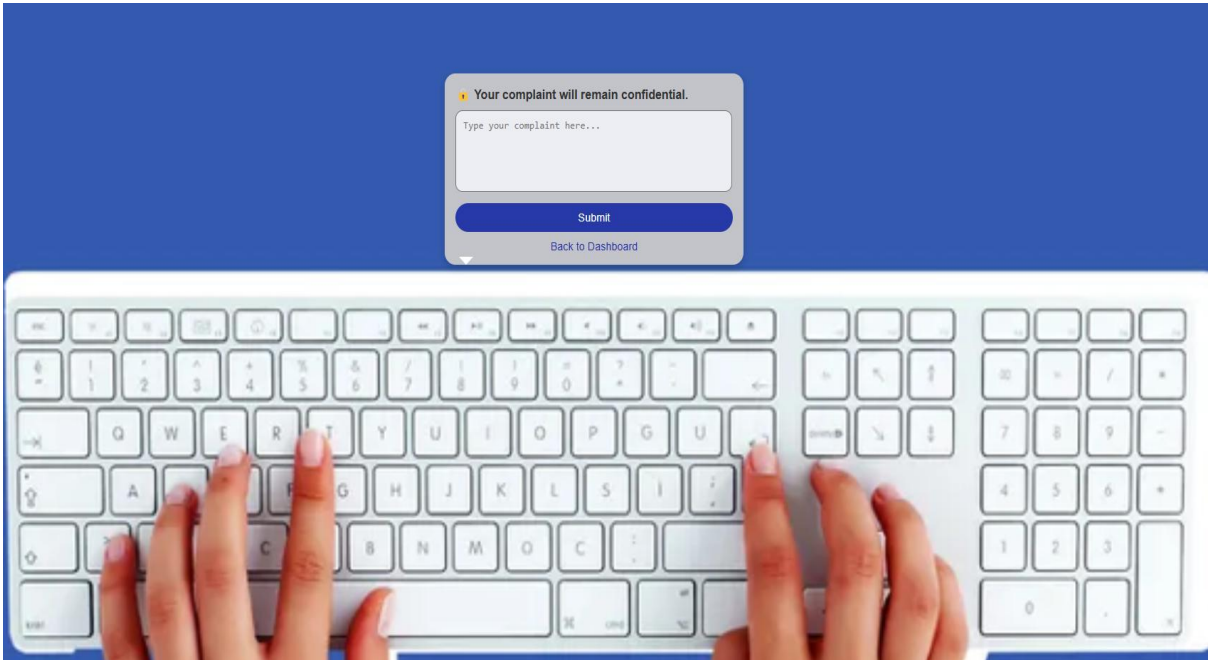
Hello Student

Book a Room

You have not booked a room yet.

Room Number	Type	Fee (per semester)	Action
102	Single	₹5,000.00	Cash <button>Book & Pay</button>
201	Double	₹8,000.00	Cash <button>Book & Pay</button>
202	Double	₹8,000.00	Cash <button>Book & Pay</button>

[Back to Dashboard](#)

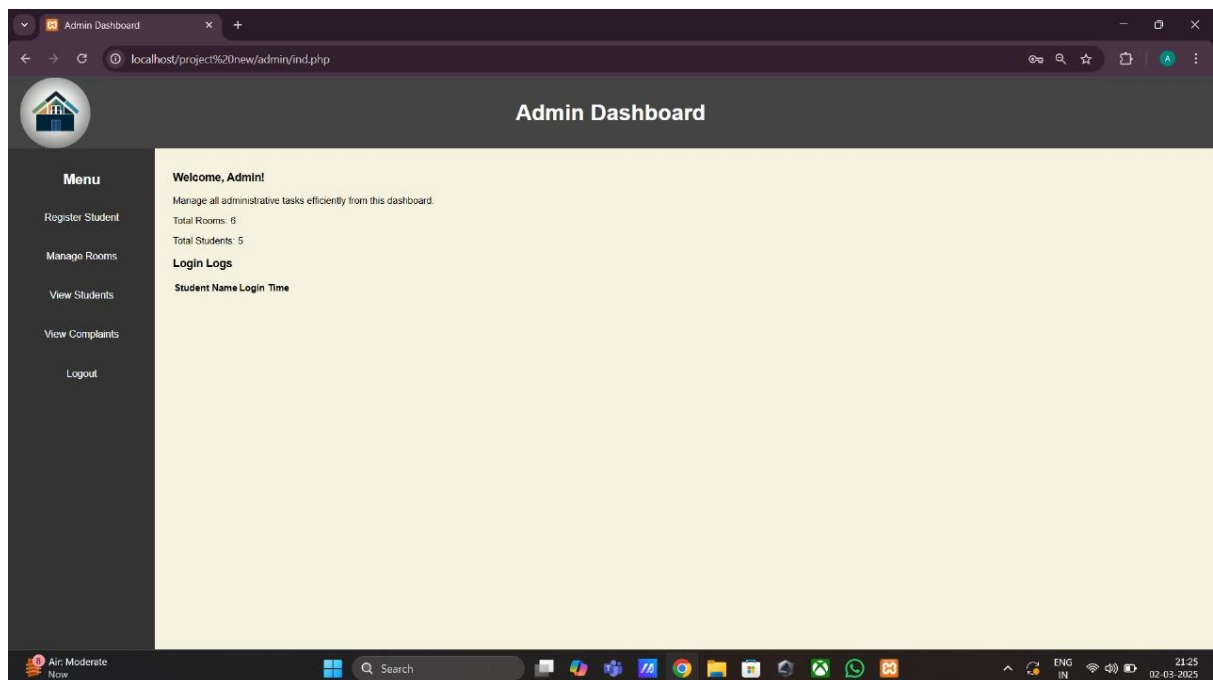


Admin Module :

Admin Login

Login

[Click here for Student Login](#)





Register New Student



Name:

Email:

Password:

Room Number:

Payment Amount:

Payment Method:

Register



Room Management

Add New Room

Room Number:

Room Type:

Single

Room Fee (per semester):

Add Room

All Rooms

Room Number	Type	Fee	Status
101	Single	₹5,000.00	Booked
102	Single	₹5,000.00	Available
201	Double	₹8,000.00	Available
202	Double	₹8,000.00	Available

Booked Rooms

Room Number	Student Name	Student Email
101	ashish	abc@gmail.com

Back to Admin Panel



Student Complaints

Complaint ID	Student Name	Complaint	Status	Date	Actions
1	ashish	vjfvdfk	Resolved	2025-01-29 17:53:44	
2	ashish	vjfvdfk	Resolved	2025-01-29 17:58:47	
3	ashish	vjfvdfk	Pending	2025-01-29 17:58:50	Mark Resolved

Back to Admin Panel

Implementation Details:

Software/hardware specifications

For running the developed system, the following specification of the Hardware and Software including the operating system is required.

Server Side

Hardware specification:

- 1.Processor : Intel (R) Core(TM) i5-10210U
- 2.Ram : 8 Gb
- 3.Hard Disk : 1 TB

Software Specifications

Operating System: Windows ;
Front End Software: HTML, CSS, PHP;
Back End Software: PHP, MySQL

Client Side

An internet browser's like Firefox, Google Chrome.

Future Scope:

- Logical and technical error will be less found in future.
- System will be more user friendly as compare to today's system.
- System will have more detail information about the different places and other new routes.
- In the security vision the user name and password authentication is more secure than today's model.

Conclusion

The **Hostel Management System** has been successfully developed to streamline and automate various hostel operations, including **room allocation, hostel facilities management, and complaint handling**. This system enhances efficiency by minimizing manual efforts, reducing errors, and ensuring a more organized approach to hostel administration.

With the **room allocation module**, the system ensures fair and systematic allotment of rooms based on predefined criteria, making the process seamless for both students and administrators. The **hostel facilities management** feature helps in maintaining and monitoring essential services, ensuring a comfortable living environment for residents.

Additionally, the **complaint management module** provides an effective platform for students to report issues, track their resolution status, and ensure prompt responses from the administration.

Overall, the **Hostel Management System** improves transparency, enhances user experience, and contributes to the smooth functioning of hostel operations. With further enhancements, such as real-time notifications and predictive maintenance, this system can continue to evolve to meet the growing needs of hostel management.

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